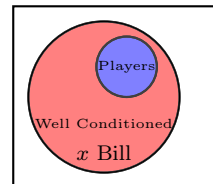


1. Construct a Venn diagram to determine the validity of the given argument.

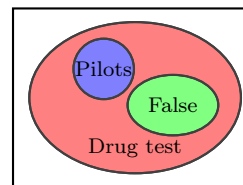
- a. 1. All basketball players are well conditioned.
 2. Bill is well-conditioned.

 Therefore, Bill is a basketball player.
Invalid



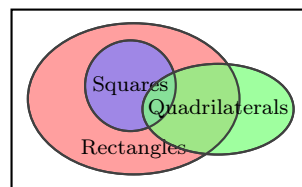
- b. 1. All pilots must take a drug test.
 2. Some people who take drug tests get false positives.

 Therefore, some pilots get false-positives.
Invalid



- c. 1. All squares are rectangles.
 2. Some quadrilaterals are squares.

 Therefore, some quadrilaterals are rectangles.
Valid



2. Using the symbolic representations

b : I ride a bicycle to school.

s : It is snowing.

r : It is raining.

express the following statements in symbolic form.

- a. It is raining and I rode my bicycle to school.
 $r \wedge b$
- b. If it is raining or snowing, then I will not ride my bicycle to school.
 $(r \vee s) \rightarrow \sim b$

3. Let x be an integer. Using the symbolic representations

p : x is a prime number.

o : x is odd.

t : x is equal to 2.

e : x is divisible by 8.

express the following in words.

- a. $p \rightarrow (o \vee t)$
 If $\underbrace{x \text{ is a prime number}}_p$ then $\underbrace{x \text{ is odd}}_o$ or $\underbrace{x \text{ is equal to 2}}_t$.
- b. $e \rightarrow \sim p$
 If $\underbrace{x \text{ is divisible by 8}}_t$ then $\underbrace{x \text{ is not prime}}_{\sim p}$.
- c. $\sim o \vee \sim e$
 If $\underbrace{x \text{ is not odd}}_{\sim o}$ or $\underbrace{x \text{ is not even}}_{\sim e}$.
- d. $\sim (o \wedge e) \wedge t$
 It is not the case that $\underbrace{x \text{ is odd}}_o$ and $\underbrace{x \text{ is even}}_e$, and $\underbrace{x \text{ is equal to 2}}_t$.

4. Construct a truth table for the symbolic expression.

a. $p \wedge (q \rightarrow \sim p)$

p	q	$\sim p$	$q \rightarrow \sim p$	$p \wedge (q \rightarrow \sim p)$
T	T	F	F	F
T	F	F	T	T
F	T	T	T	F
F	F	T	T	F

b. $(x \rightarrow y) \vee \sim x$

x	y	$x \rightarrow y$	$\sim x$	$(x \rightarrow y) \vee \sim x$
T	T	T	F	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

c. $\sim a \wedge (b \rightarrow c)$

a	b	c	$\sim a$	$b \rightarrow c$	$\sim a \wedge (b \rightarrow c)$
T	T	T	F	T	F
T	T	F	F	F	F
T	F	T	F	T	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	T	T

5. Symbolically write the negation of the following statements.

a. The grass is not greener.

g : The grass is greener.

Statement: g

Negation: $\sim g$

b. It is not the case that, if our car is not fixed then we cannot drive to Asheville.

f : Our car is fixed.

d : We can drive to Asheville.

Statement: $\sim (\sim f \rightarrow \sim d)$

Negation: $\sim (\sim (f \rightarrow \sim d)) \equiv \sim f \rightarrow \sim d$

6. Write the negation of the following statements.

a. Some freshmen are taking Math 107.

No freshmen are taking Math 107.

b. Some seniors are not taking Math 107.

All seniors are taking Math 107.

c. All football players need to attend practice.

Some football players *do not* need to attend practice.

d. No professors attend football practice.

Some professors *do* attend football practice.

7. Using the statement, "If your order is over \$50, then you will not have to pay shipping fees." write the following:

o : The order is over \$50.

p : You have to pay shipping fees. Statement: $o \rightarrow \sim p$

a. **Converse:**

$\sim p \rightarrow o$: If $\underbrace{\sim p}_{\sim p}$ then $\underbrace{o}_{o}$

b. Inverse:

$\sim o \rightarrow \sim (\sim p)$: If $\underbrace{\text{your order is not over \$50}}_{\sim o}$ then $\underbrace{\text{you have to pay shipping fees}}_{\sim(\sim p) \equiv p}$.

c. Contrapositive:

$\sim (\sim p) \rightarrow \sim o$: If $\underbrace{\text{you have to pay shipping fees}}_{\sim(\sim p) \equiv p}$ then $\underbrace{\text{your order is not over \$50}}_{\sim o}$.

d. Contrapositive in terms of a sufficient condition:

$\sim (\sim p) \rightarrow \sim o$: $\underbrace{\text{Having to pay shipping fees is sufficient for your order to be not over \$50.}}_{\sim(\sim p) \equiv p}$

e. Negation using conjunction:

$\sim (o \rightarrow \sim p)$: Using De Morgan's law, $\sim (o \rightarrow \sim p) \equiv o \wedge \sim (\sim p) \equiv o \wedge p$
 $\underbrace{\text{Your order is over \$50}}_o$ and $\underbrace{\text{you have to pay shipping fees.}}_p$

8. Using the statement, "If the sun is going down, then the cows are coming home." write the following:

s : The sun is going down.

c : The cows are coming home. Statement: $s \rightarrow c$

a. Converse:

$c \rightarrow s$: If $\underbrace{\text{the cows are coming home}}_c$ then $\underbrace{\text{the sun is going down.}}_s$

b. Inverse:

$\sim s \rightarrow \sim c$: If $\underbrace{\text{the sun is not going down}}_{\sim s}$ then $\underbrace{\text{the cows are not coming home.}}_{\sim c}$

c. Contrapositive:

$\sim c \rightarrow \sim s$: If $\underbrace{\text{the cows are not coming home}}_{\sim c}$ then $\underbrace{\text{the sun is not going down.}}_{\sim s}$

d. Contrapositive in terms of a necessary condition:

$\sim c \rightarrow \sim s$: $\underbrace{\text{The sun not going down is necessary for the cows not coming home.}}_{\sim s}$

e. Negation using conjunction:

$\sim (c \rightarrow s)$: Using De Morgan's law, $\sim (c \rightarrow s) \equiv c \wedge \sim s$
 $\underbrace{\text{The cows are coming home}}_{\sim c}$ and $\underbrace{\text{the sun is not going down.}}_{\sim s}$

9. Use a truth table to determine if the following arguments are valid.

- a. b : You buy the bike. 1. You buy the bike or the tent.
 t : You buy the tent. 2. You buy the bike.
 Therefore, you do not buy the tent.

		①	②			Argument
b	t	$b \vee t$	b	$1 \wedge 2$	$\sim t$	$(1 \wedge 2) \rightarrow c$
T	T	T	T	T	F	F
T	F	T	T	T	T	T
F	T	T	F	F	F	T
F	F	F	F	F	T	T

The argument is not always true so it is, **invalid**

- b. r : You register early. 1. If you register early, then you will receive a discount.
 d : You receive a discount. 2. You register early.
 Therefore, you will receive a discount.

		①	②		c	Argument
r	d	$r \rightarrow d$	r	$1 \wedge 2$	r	$(1 \wedge 2) \rightarrow c$
T	T	T	T	T	T	T
T	F	F	T	F	T	T
F	T	T	F	F	F	T
F	F	T	F	F	F	T

The argument is always true so it is, **valid**

- c. b : You buy the property. 1. If you buy the property, then you will need a loan.
 n : You need a loan. 2. If you need a loan, then you will pay 8%.
 p : You will pay 8%. Therefore, if you buy the property then you will pay 8%.

			①	②		c	Argument
b	n	p	$b \rightarrow n$	$n \rightarrow p$	$1 \wedge 2$	$b \rightarrow p$	$(1 \wedge 2) \rightarrow c$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

The argument is always true so it is, **valid**

- d. 1. Nothing intelligible ever puzzles me. i : It is intelligible.
 2. Logic puzzles me. p : It puzzles me.
 Therefore, logic is unintelligible. ℓ : It is logic.

			①	②					c	Argument
i	p	ℓ	$\sim p$	$i \rightarrow \sim p$	$\ell \rightarrow p$	$1 \wedge 2$	$\sim i$	$\ell \rightarrow \sim i$	$(1 \wedge 2) \rightarrow c$	
T	T	T	F	F	T	F	F	F	T	
T	T	F	F	F	T	F	F	T	T	
T	F	T	T	T	F	F	F	F	T	
T	F	F	T	T	T	T	F	T	T	
F	T	T	F	T	T	T	T	T	T	
F	T	F	F	T	T	T	T	T	T	
F	F	T	T	T	F	F	T	T	T	
F	F	F	T	T	T	T	T	T	T	

The argument is always true so it is, **valid**